

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Annexure A

Pseudocode Writing Task (Algorithm Design)

Criteria	Problem Understanding (2)	Algorithm Design (3)	Use of Control Structures (2)	Pseudocode Structure (1)	Completeness (2)	Total(10)
Weightage	20%	30%	20%	10%	20%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

*I M MEGHANADH(192325026) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
MEGHANADH

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Annexure A

Q1. Dataset: Online Shopping Transactions

Order ID	Products Purchased
O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

Question:

Design pseudocode for the **Apriori algorithm** to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration.

Q2. Dataset: Library Book Borrowing Records

Borrow ID	Books Borrowed
B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Question:

Develop pseudocode for the **FP-Growth algorithm** to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. (BL4)

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3

{Streaming, Cloud Storage} 2

Question:

Write pseudocode to generate **association rules** from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. (BL3)

Q4. Dataset: Student Course Enrollment

Student ID Courses Enrolled

S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Question:

Design pseudocode for **constraint-based frequent itemset mining** where only itemsets containing the course "**Python**" are considered during the mining process. Explain how the constraint affects candidate generation and pruning. (BL4)

Q5. Dataset: Hospital Treatment Hierarchy

Patient ID Treatments Received

P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG
P5	Healthcare, Orthopedics

Question:

Develop pseudocode for **multilevel association rule mining** using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

Q1. Apriori Algorithm – Online Shopping Transactions

Dataset

Order Products

O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

The question asks for pseudocode that generates frequent itemsets using a specified minimum-support threshold and shows candidate generation and pruning.

Pseudocode

ALGORITHM Apriori

INPUT:

Transaction database D

Minimum support threshold minsup

OUTPUT:

All frequent itemsets

BEGIN

L1 ← Find all frequent 1-itemsets in D

k ← 2

WHILE L(k-1) is not empty DO

 Ck ← GenerateCandidates(L(k-1))

 FOR each transaction T in D DO

 FOR each candidate c in Ck DO

 IF c is subset of T THEN

 count[c] ← count[c] + 1

 END IF

 END FOR

END FOR

$L_k \leftarrow$ empty set

FOR each candidate c in C_k DO

$\text{support}[c] \leftarrow \text{count}[c] / \text{number_of_transactions}$

 IF $\text{support}[c] \geq \text{minsup}$ THEN

 Add c to L_k

 ELSE

 Remove c from consideration

 END IF

END FOR

$k \leftarrow k + 1$

END WHILE

RETURN $L_1 \cup L_2 \cup \dots \cup L_{(k-1)}$

END

Candidate Generation

FUNCTION GenerateCandidates(L_{prev})

$C \leftarrow$ empty set

 FOR each itemset p in L_{prev} DO

 FOR each itemset q in L_{prev} DO

 IF first $(k-2)$ items of p

 are equal to first $(k-2)$ items of q THEN

 candidate $\leftarrow p \text{ UNION } q$

 IF every $(k-1)$ -subset of candidate

 exists in L_{prev} THEN

Add candidate to C

END IF

END IF

END FOR

END FOR

RETURN C

END FUNCTION

Example with minimum support = 40%

There are 5 transactions.

Minimum count:

$$5 \times 40\% = 2$$

1-itemsets

Item	Count	Support
Mobile	4	80%
Earphones	3	60%
Charger	3	60%

All three are frequent.

2-itemsets

Itemset	Count	Support
Mobile, Earphones	3	60%
Mobile, Charger	2	40%
Earphones, Charger	2	40%

All are frequent.

3-itemset

Itemset	Count	Support
Mobile, Earphones, Charger	1	20%

It is **pruned** because $20\% < 40\%$.

Conclusion

Apriori reduces the search space by eliminating candidates whose support is below the minimum

threshold. The Apriori property states that **if an itemset is infrequent, all of its supersets are also infrequent.**

Q2. FP-Growth Algorithm – Library Records

Dataset

Borrow ID Books

B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

The question specifically requires FP-Growth **without generating candidate itemsets** and asks for the FP-Tree construction process.

Pseudocode

ALGORITHM FP-Growth

INPUT:

Transaction database D
Minimum support minsup

OUTPUT:

Frequent itemsets

BEGIN

Scan D and calculate support count of every item

Remove items whose support < minsup

Sort remaining items in descending
order of support

Create an empty FP-Tree

FOR each transaction T in D DO

Remove infrequent items from T

Sort items according to global frequency

Insert the ordered transaction into FP-Tree

Update node counts

END FOR

Create a Header Table linking identical items

Mine the FP-Tree recursively

RETURN all frequent itemsets

END

FP-Tree Construction

Assume minimum support = **40%**.

There are 5 transactions:

$$\text{Minimum count} = 5 \times 40\% = 2$$

Item counts:

Item	Count
Python	4
Machine Learning	4
Data Science	3

All three items are frequent.

A possible frequency order is:

Python → Machine Learning → Data Science

Reorder transactions:

B1: Python → Data Science

B2: Python → Machine Learning

B3: Machine Learning → Data Science

B4: Python → Machine Learning → Data Science

B5: Python → Machine Learning

Simplified FP-Tree

ROOT

```
|
+-- Python:4
|  |
|  +-- Data Science:1
|  |
|  +-- Machine Learning:3
|      |
|      +-- Data Science:1
|
+-- Machine Learning:1
    |
    +-- Data Science:1
```

FP-Growth Mining

FP-Growth creates conditional pattern bases for each item and recursively mines them.

For example:

Machine Learning

It appears in B2, B3, B4 and B5.

Its support:

$$\frac{4}{5} = 80\%$$

Python + Machine Learning

Appears in B2, B4 and B5.

$$Support = \frac{3}{5} = 60\%$$

Machine Learning + Data Science

Appears in B3 and B4.

$$Support = \frac{2}{5} = 40\%$$

Python + Data Science

Appears in B1 and B4.

$$Support = \frac{2}{5} = 40\%$$

Frequent Itemsets

At 40% support:

- {Python} → 80%
- {Machine Learning} → 80%
- {Data Science} → 60%

- {Python, Machine Learning} → 60%
- {Python, Data Science} → 40%
- {Machine Learning, Data Science} → 40%

The main advantage is that **FP-Growth avoids repeatedly generating candidate itemsets**, making it efficient for large databases.

Q3. Association Rule Generation

The given frequent combinations are:

Itemset	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Pseudocode

ALGORITHM GenerateAssociationRules

INPUT:

Frequent itemsets F

Minimum confidence minconf

OUTPUT:

Valid association rules

BEGIN

FOR each frequent itemset I in F DO

IF $\text{size}(I) \geq 2$ THEN

Generate all non-empty proper subsets
of I

FOR each subset A of I DO

$B \leftarrow I - A$

confidence $\leftarrow$

Support(I) / Support(A)

IF confidence $\geq$ minconf THEN

Generate rule:

$A \rightarrow B$

Output rule, support, confidence

END IF

END FOR

END IF

END FOR

END

Example

Consider:

$\{I, nternetStreaming\}$

Support count = 3.

For:

Internet $\rightarrow$ Streaming

$$Confidence = \frac{Support(I, nternetStreaming)}{Support(Internet)}$$

$$= \frac{3}{4} = 75\%$$

For:

Streaming $\rightarrow$ Internet

$$Confidence = \frac{3}{4} = 75\%$$

Therefore, if minimum confidence is **70%**, both rules are accepted.

Streaming + Cloud Storage

Support count = 2.

Streaming → Cloud Storage

$$\text{Confidence} = \frac{2}{4} = 50\%$$

Cloud Storage → Streaming

$$\text{Confidence} = \frac{2}{3} = 66.67\%$$

If minimum confidence = 70%, both rules are rejected.

Final

With **minimum confidence = 70%**:

Rule	Confidence	Result
Internet → Streaming	75%	Accepted
Streaming → Internet	75%	Accepted
Streaming → Cloud Storage	50%	Rejected
Cloud Storage → Streaming	66.67%	Rejected

Q4. Constraint-Based Frequent Itemset Mining

Dataset

Student Courses

S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Constraint:

Only itemsets containing Python are considered.

Pseudocode

ALGORITHM ConstraintBasedMining

INPUT:

Transaction database D
Minimum support minsup
Required item = Python

OUTPUT:

Frequent itemsets containing Python

BEGIN

$C1 \leftarrow \{\text{Python}\}$

Calculate support(Python)

IF support(Python) $\geq$ minsup THEN

 Add {Python} to L1

END IF

$k \leftarrow 2$

WHILE L(k-1) is not empty DO

 Generate candidate itemsets Ck

 FOR each candidate c in Ck DO

 IF Python is NOT in c THEN

 Prune c

 CONTINUE

 END IF

 Calculate support(c)

 IF support(c) $\geq$ minsup THEN

 Add c to Lk

 ELSE

 Prune c

 END IF

 END FOR

$k \leftarrow k + 1$

END WHILE

RETURN all frequent itemsets

END

Apply the Constraint

Python occurs in:

- S1
- S2
- S4
- S5

Therefore:

$$\text{Support}(\text{Python}) = \frac{4}{5} = 80\%$$

Python + Database

Appears in S1 and S4.

$$\text{Support} = \frac{2}{5} = 40\%$$

Python + AI

Appears in S2 and S4.

$$\text{Support} = \frac{2}{5} = 40\%$$

Python + Database + AI

Appears only in S4.

$$\text{Support} = \frac{1}{5} = 20\%$$

If minimum support = 40%

Frequent itemsets satisfying the Python constraint:

Itemset	Support
{Python}	80%
{Python, Database}	40%
{Python, AI}	40%

{Python, Database, AI} is pruned because its support is only 20%.

Effect of Constraint

The Python constraint reduces the search space because itemsets such as:

- {Database}
- {AI}

- {Database, AI}

do not need to be considered.

Thus, fewer candidates are generated, reducing computation time and memory usage.

Q5. Multilevel Association Rule Mining

Dataset

Patient Treatments

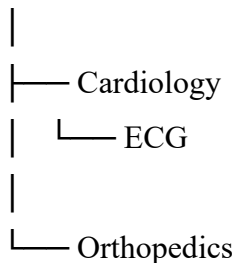
P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG
P5	Healthcare, Orthopedics

The question asks for multilevel association mining using the treatment hierarchy.

Treatment Hierarchy

A simple hierarchy from the dataset is:

Healthcare



Pseudocode

ALGORITHM MultilevelAssociationMining

INPUT:

Transaction database D
Treatment hierarchy H
Minimum support minsup

OUTPUT:

Association rules at different hierarchy levels

BEGIN

FOR each hierarchy level L DO

Map detailed treatments to their
corresponding higher-level concepts

Generate transactions at level L

Apply Apriori or FP-Growth

Find frequent itemsets

Generate association rules

Store rules for level L

END FOR

Return rules from all hierarchy levels

END

Level 1 – General Treatment

At the highest level:

Healthcare

occurs in all 5 patients.

$$Support(Healthcare) = \frac{5}{5} = 100\%$$

So Healthcare is highly frequent.

Level 2 – Treatment Categories

Cardiology

Cardiology appears in:

- P1
- P3
- P4

$$Support(Cardiology) = \frac{3}{5} = 60\%$$

Orthopedics

Orthopedics appears in:

- P2
- P5

$$\text{Support}(\text{Orthopedics}) = \frac{2}{5} = 40\%$$

Level 3 – Specific Treatment

ECG

ECG appears in:

- P1
- P4

$$\text{Support}(\text{ECG}) = \frac{2}{5} = 40\%$$

Example Association Rules

Cardiology → Healthcare

All patients having Cardiology also have Healthcare.

$$\begin{aligned} \text{Confidence} &= \frac{\text{Support}(C, \text{cardiologyHealthcare})}{\text{Support}(\text{Cardiology})} \\ &= \frac{60\%}{60\%} = 100\% \end{aligned}$$

Therefore:

Cardiology → Healthcare

- Support = **60%**
- Confidence = **100%**

Orthopedics → Healthcare

$$\text{Support} = 40\%$$

$$\text{Confidence} = \frac{40\%}{40\%} = 100\%$$

Therefore:

Orthopedics → Healthcare

- Support = **40%**
- Confidence = **100%**

ECG → Cardiology

ECG occurs in P1 and P4, and both also have Cardiology.

$$\text{Support}(E, \text{CGCardiology}) = \frac{2}{5} = 40\%$$

$$\text{Confidence}(\text{ECG} \rightarrow \text{Cardiology}) = \frac{40\%}{40\%} = 100\%$$

Therefore:

ECG → Cardiology

- Support = **40%**
- Confidence = **100%**

Why Concept Hierarchies Are Useful

Concept hierarchies allow patterns to be discovered at different levels of abstraction.

For example:

Healthcare



Cardiology



ECG

At the general level, we discover broad treatment patterns. At lower levels, we discover more specific relationships.

This helps because:

1. **High-level patterns** provide a general overview.
2. **Lower-level patterns** provide detailed information.
3. Rare detailed patterns can be studied through their higher-level categories.
4. It supports better decision-making and knowledge discovery.
5. It allows users to move from general patterns to specific patterns.

Final Answer

Multilevel association mining begins with higher-level concepts such as **Healthcare**, then moves to **Cardiology/Orthopedics**, and finally to specific treatments such as **ECG**. This hierarchical approach provides both general and detailed treatment relationships.